

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

**COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN**  
**COURSE CODE: CSA1101**

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### **COURSE OUTCOME COVERED**

***CO4:***

implement object-oriented system layers using design principles and patterns, and integrate access and view layers with OODBMS (*BL5*)

*Assessment Tool Weightage: 40%*

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**QUESTIONS: 5**

**Total Marks:40**

Annexure A

### **Industry Problem Solving Task**

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***Code of Conduct:***

*I Mathusri P(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** Mathusri P

## Annexure A

# Industry Problem Solving Task

## 1. Smart Banking and Financial Management System

A financial institution develops a **Smart Banking and Financial Management System** to manage customer accounts, fund transfers, loan processing, bill payments, and transaction history. The application follows a layered object-oriented architecture consisting of a presentation layer, service layer, domain layer, and data access layer connected to an OODBMS.

The system uses the **Factory Pattern** to create different account types, the **Strategy Pattern** for loan-interest and payment calculation policies, and the **Observer Pattern** for transaction and fraud alerts. A **Repository Pattern** is used to manage communication with persistent banking objects. The architecture is designed to support future services such as investment management and digital wallets.

### Question:

Analyze how object-oriented principles, layered architecture, and design patterns contribute to the scalability, security, flexibility, and maintainability of the banking system. Explain how the application layers communicate with the OODBMS to store and retrieve persistent banking objects.

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## 2. Smart Retail and Inventory Management System

A retail organization develops a **Smart Retail and Inventory Management System** to manage products, suppliers, customers, purchase orders, sales transactions, and warehouse stock. The system follows a layered object-oriented architecture with presentation, business, domain, and data access layers connected to an OODBMS.

The system applies the **MVC Pattern** for managing user interfaces, the **Factory Pattern** for creating different product categories, and the **Observer Pattern** to notify managers when inventory reaches a minimum threshold. The **DAO Pattern** handles communication with persistent inventory objects. The architecture allows future integration of automated stock prediction and robotic warehouse management.

### Question:

Evaluate how object-oriented design principles, layered architecture, and design patterns improve the flexibility, scalability, and maintainability of the retail system. Explain how the different layers interact with the OODBMS for efficient inventory persistence and retrieval.

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### 3. Smart Hotel and Hospitality Management System

A hotel chain develops a **Smart Hotel Management System** to manage guest registration, room reservations, check-in/check-out, payments, room services, and customer feedback. The application uses a layered object-oriented architecture consisting of presentation, service, domain, and data access layers connected to an OODBMS.

The system implements the **Strategy Pattern** for different pricing and discount policies, the **Factory Pattern** for creating room and service objects, and the **Observer Pattern** for reservation and service notifications. A **Repository Pattern** is used to simplify communication between business components and persistent hotel objects. Future enhancements include smart-room automation and personalized guest services.

#### Question:

Analyze how object-oriented principles and design patterns support flexibility, reusability, and maintainability in the hotel management system. Explain how the layered architecture coordinates reservation processing and communicates with the OODBMS for persistent data management.

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### 4. Smart Logistics and Delivery Management System

A logistics company develops a **Smart Logistics and Delivery Management System** to manage shipment registration, warehouse operations, vehicle allocation, delivery tracking, and customer notifications. The system adopts a layered object-oriented architecture with presentation, business, domain, and data access layers connected to an OODBMS.

The system uses the **Strategy Pattern** for selecting optimal delivery routes, the **Factory Pattern** for creating different shipment types, and the **Observer Pattern** for real-time delivery status notifications. A **DAO Pattern** manages persistent shipment and vehicle information. The architecture is designed to support future integration with autonomous delivery vehicles and AI-based route optimization.

#### Question:

Examine how object-oriented design principles, layered architecture, and design patterns improve the adaptability, scalability, and maintainability of the logistics system. Explain how the layers communicate with the OODBMS to manage persistent shipment, vehicle, and delivery information.

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### 5. Smart Energy Management System

An energy management company develops a **Smart Energy Management System** to monitor electricity consumption, manage smart meters, analyze energy usage, and generate alerts for abnormal consumption. The system follows a layered object-oriented architecture where the

presentation layer provides dashboards, the business layer performs energy analysis, the domain layer manages energy-related rules, and the data access layer communicates with an OODBMS.

The system uses the **Observer Pattern** for consumption alerts, the **Strategy Pattern** for different energy optimization algorithms, and the **Factory Pattern** for creating different smart-meter objects. A **Repository Pattern** is used to manage persistent energy data. The architecture is designed to accommodate future renewable-energy monitoring and AI-based consumption prediction.

**Question:**

Evaluate how object-oriented principles, layered architecture, and design patterns contribute to the scalability, extensibility, and maintainability of the energy management system. Explain the interaction between the system layers and the OODBMS for storing, retrieving, and managing persistent energy-monitoring objects.

## RUBRICS FOR CALCULATION

### Industry Problem Solving Task

| Criteria                            | Weightage | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                                |
|-------------------------------------|-----------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|----------------------------------------------------|
| Understanding of Industry Problem   | 20%       | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding ; misinterprets problem context |
| Application of Engineering Concepts | 25%       | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts            |
| Solution Design                     | 25%       | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                    |
| Use of Tools                        | 15%       | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                  |
| Analysis                            | 10%       | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                                |
| Professionalism                     | 5%        | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation                |

## **Q1. Smart Banking and Financial Management System**

### **Introduction**

The Smart Banking and Financial Management System is designed to manage customer accounts, fund transfers, loan processing, bill payments, and transaction history. Since banking applications handle sensitive financial information, the system requires strong security, reliable data persistence, scalability, and maintainability. A layered object-oriented architecture with design patterns provides a suitable solution.

### **1. Object-Oriented Principles**

The system applies the major object-oriented principles of **encapsulation, abstraction, inheritance, and polymorphism**.

**Encapsulation** protects sensitive banking data such as account balance, PIN, and transaction details by restricting direct access. Operations such as deposit, withdrawal, and transfer are performed through controlled methods.

**Abstraction** hides complex implementation details from users. For example, a customer only requests a fund transfer without knowing how the transaction is processed internally.

**Inheritance** can be used to create different account types from a common Account class. SavingsAccount, CurrentAccount, and SalaryAccount can reuse common account properties while providing specialized behavior.

**Polymorphism** allows different account types or payment methods to implement the same operation differently. This makes the system flexible and extensible.

### **2. Layered Architecture**

The system contains four major layers.

#### **Presentation Layer**

This layer provides the banking interface through which customers perform login, balance enquiry, fund transfer, loan requests, and bill payments. It is responsible only for displaying information and collecting user input.

#### **Service Layer**

The Service Layer coordinates application use cases such as fund transfer, loan processing, bill payment, and transaction management. It controls the sequence of operations without containing low-level database code.

#### **Domain Layer**

This layer contains the core banking objects and business rules. Important classes include:

- Customer
- Account
- SavingsAccount
- CurrentAccount
- Loan
- Transaction
- Bill
- Payment

The domain layer validates rules such as minimum balance, transfer limits, loan eligibility, and payment conditions.

#### **Data Access Layer**

This layer communicates with the OODBMS through repositories. It stores and retrieves persistent objects such as Customer, Account, Loan, and Transaction objects.

### **3. Application of Design Patterns**

#### **Factory Pattern**

The Factory Pattern creates appropriate account objects without exposing object-creation logic to the client.

For example, when a customer selects a Savings Account, the factory creates a SavingsAccount object. Similarly, it can create CurrentAccount or SalaryAccount objects.

This makes the system easier to extend when new account types are introduced.

#### **Strategy Pattern**

The Strategy Pattern is used for loan-interest calculation and payment policies.

Different strategies can be created for:

- Home loan interest
- Education loan interest
- Personal loan interest
- Different payment methods

A new calculation policy can be added without modifying the main loan-processing class.

#### **Observer Pattern**

The Observer Pattern is useful for transaction and fraud alerts.

When a transaction occurs, interested components such as notification services and fraud-monitoring services can automatically receive the event.

This supports real-time notifications without tightly coupling the transaction class with every notification mechanism.

#### **Repository Pattern**

The Repository Pattern provides an abstraction for storing and retrieving persistent banking objects. Services communicate with repositories instead of directly interacting with the OODBMS.

### **4. Communication with OODBMS**

The Service Layer requests the required operation from the Repository. The Repository communicates with the OODBMS to store or retrieve persistent objects.

For example, during a fund transfer:

1. The customer submits a transfer request.
2. The Presentation Layer sends the request to the Service Layer.
3. The Service Layer validates the transaction.
4. The Domain Layer applies banking rules.
5. The Repository retrieves the required Account objects.
6. The transaction is processed.
7. Updated Account and Transaction objects are stored in the OODBMS.
8. The Observer mechanism generates transaction or fraud notifications.
9. The result is returned to the customer.

Object identity and relationships between customers, accounts, loans, and transactions can be maintained directly through an OODBMS.

### **5. Security**

Security is improved by separating responsibilities between layers. Sensitive data is not directly exposed to the Presentation Layer. Authentication and authorization can be applied before accessing banking services.

Encapsulation protects account information, while service-level validation prevents unauthorized or invalid transactions.

## **6. Scalability and Flexibility**

The architecture supports scalability because individual components can be improved independently. New services such as investment management and digital wallets can be added as separate domain and service components.

Design patterns reduce dependencies and allow new account types, payment policies, and notification mechanisms to be introduced without major changes to existing code.

## **Conclusion**

The combination of object-oriented principles, layered architecture, Factory, Strategy, Observer, and Repository patterns provides a secure, flexible, scalable, and maintainable banking system. The Repository Layer provides controlled communication with the OODBMS, allowing banking objects to be stored and retrieved efficiently while maintaining object relationships and persistence.

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## **Q2. Smart Retail and Inventory Management System**

### **Introduction**

The Smart Retail and Inventory Management System manages products, suppliers, customers, purchase orders, sales transactions, and warehouse stock. Since retail systems handle continuously changing inventory and transaction information, the architecture must support efficient data access, flexibility, scalability, and maintainability.

### **1. Object-Oriented Design Principles**

**Encapsulation** protects product prices, stock quantities, supplier information, and transaction details. These values are modified through controlled methods.

**Abstraction** hides database and inventory-processing complexity from users. A manager can request stock information without knowing how the data is retrieved from the OODBMS.

**Inheritance** can be used to represent different product categories using a common Product class.

**Polymorphism** allows different product types to implement category-specific operations while following a common interface.

These principles improve reuse and reduce unnecessary dependencies.

### **2. Layered Architecture**

#### **Presentation Layer**

The Presentation Layer provides interfaces for customers, warehouse employees, managers, and administrators. It supports product search, order placement, stock monitoring, and sales reporting.

#### **Business Layer**

The Business Layer processes retail operations such as order validation, stock checking, purchasing, sales processing, and inventory threshold checking.

#### **Domain Layer**

The Domain Layer contains important objects such as:

- Product
- Supplier
- Customer
- PurchaseOrder



- SalesTransaction
- Warehouse
- StockItem

Business rules related to inventory and sales are implemented within this layer.

### **Data Access Layer**

The Data Access Layer communicates with the OODBMS and manages persistent retail objects.

### **3. MVC Pattern**

The MVC Pattern separates the user interface into Model, View, and Controller components.

The **Model** represents product and inventory data.

The **View** displays products, stock information, and transaction details.

The **Controller** receives user requests and coordinates the required operations.

This separation makes the user interface easier to modify without changing the underlying business logic.

### **4. Factory Pattern**

The Factory Pattern creates different product objects according to their category.

For example, the factory can create:

- ElectronicProduct
- GroceryProduct
- ClothingProduct
- FurnitureProduct

The client does not need to know the exact creation process.

This makes the system extensible when new product categories are added.

### **5. Observer Pattern**

The Observer Pattern is used for inventory threshold notifications.

When the quantity of a product reaches the minimum level, the stock object can notify:

- Store Manager
- Purchase Department
- Warehouse Manager
- Notification Service

This allows automatic alerts without tightly coupling the stock class to every notification component.

### **6. DAO Pattern and OODBMS**

The DAO Pattern separates database operations from business logic.

For example, a ProductDAO or InventoryDAO can provide operations such as:

- saveProduct()
- findProduct()
- updateStock()
- deleteProduct()
- findLowStockItems()

The DAO communicates with the OODBMS while the Business Layer uses the DAO interface.

When stock is updated, the updated Product or StockItem object is persisted in the OODBMS.

When a manager requests inventory information, the DAO retrieves the required persistent objects.

### **7. Scalability and Future Development**

The architecture can support automated stock prediction by adding a separate prediction service. Robotic warehouse management can also be integrated through infrastructure services without modifying the core inventory classes.

Low coupling makes it easier to integrate external systems, while high cohesion keeps individual components focused on specific responsibilities.

## **8. Analysis**

The combination of MVC, Factory, Observer, and DAO provides a flexible architecture. MVC separates presentation concerns, Factory simplifies object creation, Observer supports real-time stock alerts, and DAO separates persistence operations.

The OODBMS is particularly useful for maintaining relationships between products, suppliers, warehouses, purchase orders, and sales transactions.

## **Conclusion**

The Smart Retail and Inventory Management System achieves flexibility, scalability, and maintainability through object-oriented principles and appropriate design patterns. The layered structure ensures that presentation, business processing, domain rules, and persistence remain separated, while the DAO provides efficient communication with the OODBMS.

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## **Q3. Smart Hotel and Hospitality Management System**

### **Introduction**

The Smart Hotel Management System manages guest registration, room reservations, check-in, check-out, payments, room services, and customer feedback. A flexible architecture is required because hotel operations involve many interacting objects and may require future smart-room and personalized-service features.

### **1. Object-Oriented Principles**

**Encapsulation** protects guest information, payment information, room status, and reservation details.

**Abstraction** allows users to perform hotel operations without knowing the internal processing.

**Inheritance** can represent different room types such as StandardRoom, DeluxeRoom, and SuiteRoom.

**Polymorphism** allows different room types and services to apply different pricing or service behavior.

These principles improve reusability and reduce code duplication.

### **2. Layered Architecture**

#### **Presentation Layer**

The Presentation Layer provides interfaces for guests, receptionists, administrators, and managers. It supports room search, reservation, check-in, payment, and feedback.

#### **Service Layer**

The Service Layer coordinates use cases such as:

- Room reservation
- Guest registration
- Check-in
- Check-out
- Payment
- Room service

#### **Domain Layer**

The Domain Layer contains objects such as:

- Guest
- Room
- Reservation
- Payment
- RoomService
- Feedback

It contains hotel-specific rules such as room availability, reservation validity, pricing, and cancellation policies.

### **Data Access Layer**

This layer communicates with the OODBMS to store and retrieve persistent hotel objects.

### **3. Strategy Pattern**

The Strategy Pattern supports different pricing and discount policies.

For example:

- Normal pricing
- Weekend pricing
- Seasonal pricing
- Loyalty discount
- Promotional discount

The Reservation Service can select the appropriate strategy based on the booking conditions.

This prevents large conditional statements and allows new pricing strategies to be added easily.

### **4. Factory Pattern**

The Factory Pattern creates different room and service objects.

For example, a RoomFactory can create StandardRoom, DeluxeRoom, and SuiteRoom objects.

A ServiceFactory can create room cleaning, food delivery, laundry, and other service objects.

This centralizes object creation and improves maintainability.

### **5. Observer Pattern**

The Observer Pattern can be used for reservation and service notifications.

When a reservation is confirmed, the notification service can automatically inform the guest.

Similarly, room-service status changes can trigger notifications.

This provides real-time communication without tightly coupling reservation processing with notification services.

### **6. Repository Pattern and OODBMS**

The Repository Pattern provides a simple interface for accessing persistent objects.

Examples include:

- GuestRepository
- RoomRepository
- ReservationRepository
- PaymentRepository

When a guest makes a reservation, the Service Layer validates the request. The Repository retrieves the selected Room object from the OODBMS and checks its availability.

After successful reservation, the Reservation object is stored and the Room object's status is updated.

The OODBMS maintains persistent relationships between guests, rooms, reservations, payments, and services.

## 7. Reservation Processing

The reservation process begins when the guest selects a room. The Presentation Layer sends the request to the Service Layer.

The Service Layer requests the Room Repository to retrieve the room. The Domain Layer checks availability and applies the appropriate pricing strategy.

If the room is available, a Reservation object is created and stored through the Repository. The room status is updated and the Observer Pattern generates a reservation confirmation.

## 8. Future Enhancement

Smart-room automation can be integrated through additional infrastructure services. Personalized guest services can use customer preferences stored as persistent objects.

The architecture allows these features to be added without changing the core reservation functionality.

## Conclusion

Object-oriented principles and design patterns provide a flexible and reusable design for the Smart Hotel Management System. Strategy supports changing pricing policies, Factory simplifies object creation, Observer provides notifications, and Repository manages persistence. The layered architecture separates responsibilities and makes the system easier to maintain and extend.

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## Q4. Smart Logistics and Delivery Management System

### Introduction

The Smart Logistics and Delivery Management System manages shipment registration, warehouse operations, vehicle allocation, delivery tracking, and customer notifications. The system must handle large numbers of shipments and provide reliable real-time tracking.

### 1. Object-Oriented Principles

**Encapsulation** protects shipment information, vehicle details, customer information, and delivery status.

**Abstraction** hides complex route selection and tracking operations from users.

**Inheritance** can represent different shipment types using a common Shipment class.

**Polymorphism** allows different shipment types to use different processing rules.

These principles improve code reuse and make the system easier to modify.

### 2. Layered Architecture

#### Presentation Layer

Provides interfaces for customers, warehouse staff, logistics managers, and administrators. It displays shipment status, delivery information, and vehicle allocation.

#### Business Layer

Handles shipment validation, vehicle allocation, warehouse processing, route selection, and delivery management.

#### Domain Layer

Contains classes such as:

- Shipment
- Vehicle
- Warehouse
- Delivery
- Customer

- Route
- Notification

The domain layer implements logistics-specific business rules.

### **Data Access Layer**

The Data Access Layer communicates with the OODBMS and manages persistent shipment, vehicle, warehouse, and delivery objects.

### **3. Strategy Pattern**

The Strategy Pattern is used for route selection.

Different route strategies can include:

- Shortest route
- Fastest route
- Lowest-cost route
- Traffic-aware route
- AI-based optimized route

The system can select an appropriate strategy depending on delivery requirements.

This allows AI-based route optimization to be added later without modifying the main delivery service.

### **4. Factory Pattern**

The Factory Pattern creates different shipment types.

For example:

- StandardShipment
- ExpressShipment
- FragileShipment
- RefrigeratedShipment

The Factory determines which object should be created based on shipment requirements.

This improves flexibility and centralizes object creation.

### **5. Observer Pattern**

The Observer Pattern supports real-time delivery notifications.

When the status of a shipment changes, observers such as:

- Customer Notification Service
- Mobile Application
- Email Service
- Logistics Dashboard

can automatically receive updates.

This avoids directly connecting the Shipment object with every notification mechanism.

### **6. DAO Pattern and OODBMS**

The DAO Pattern manages persistence operations.

Examples include:

- ShipmentDAO
- VehicleDAO
- WarehouseDAO
- DeliveryDAO

The DAO can retrieve shipment information, update delivery status, save vehicle information, and maintain warehouse records.

The OODBMS stores persistent objects and their relationships. For example, a Shipment can be associated with a Customer, Vehicle, Warehouse, and Delivery object.

## **7. Delivery Processing**

The customer registers a shipment through the Presentation Layer. The Business Layer validates the shipment details and requests the Domain Layer to create the appropriate shipment object.

The Factory creates the required shipment type. The system checks vehicle availability and selects a route using the appropriate Strategy.

The shipment and vehicle information are stored through the DAO in the OODBMS. During transportation, changes in delivery status trigger Observer notifications.

## **8. Scalability and Adaptability**

The system can support autonomous delivery vehicles by adding new vehicle classes and services. AI-based route optimization can be introduced as a new Strategy.

Because the architecture has low coupling, these additions can be made without rewriting the existing shipment-processing system.

## **Conclusion**

The Smart Logistics and Delivery Management System achieves adaptability and scalability through object-oriented principles and design patterns. Strategy enables flexible route selection, Factory supports multiple shipment types, Observer provides real-time notifications, and DAO manages persistent objects. The layered architecture ensures clear separation of responsibilities and supports future technological enhancements.

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## **Q5. Smart Energy Management System**

### **Introduction**

The Smart Energy Management System monitors electricity consumption, manages smart meters, analyzes energy usage, and generates alerts for abnormal consumption. The system must efficiently manage large volumes of monitoring data and support future renewable-energy and AI-based prediction features.

### **1. Object-Oriented Principles**

**Encapsulation** protects meter readings, energy consumption data, customer information, and alert conditions.

**Abstraction** hides the complex calculations used for energy analysis and optimization.

**Inheritance** can be used to represent different types of smart meters from a common SmartMeter class.

**Polymorphism** allows different smart-meter types and optimization algorithms to implement common operations in different ways.

These principles improve reusability and maintainability.

### **2. Layered Architecture**

#### **Presentation Layer**

The Presentation Layer provides dashboards showing:

- Current energy consumption
- Historical usage
- Meter status
- Abnormal consumption alerts
- Energy-saving recommendations

## **Business Layer**

The Business Layer performs energy analysis and coordinates operations such as consumption calculation, threshold checking, alert generation, and optimization.

## **Domain Layer**

The Domain Layer manages energy-related objects and rules.

Important classes include:

- SmartMeter
- EnergyReading
- Consumer
- EnergyUsage
- Alert
- OptimizationPolicy

## **Data Access Layer**

The Data Access Layer manages persistent energy objects and communicates with the OODBMS through repositories.

## **3. Observer Pattern**

The Observer Pattern is used for abnormal consumption alerts.

When a smart meter detects unusual consumption, the monitoring system generates an event. Observers such as the customer dashboard, mobile application, and notification service can receive the alert.

This allows multiple notification mechanisms to respond to the same event without tightly coupling them to the SmartMeter class.

## **4. Strategy Pattern**

The Strategy Pattern supports different energy optimization algorithms.

Possible strategies include:

- Peak-hour reduction
- Cost-based optimization
- Consumption balancing
- Renewable-energy optimization
- AI-based energy prediction

A suitable strategy can be selected according to the customer's requirements.

New algorithms can be introduced without changing the main energy management service.

## **5. Factory Pattern**

The Factory Pattern creates different types of smart-meter objects.

For example:

- ResidentialSmartMeter
- IndustrialSmartMeter
- CommercialSmartMeter
- SolarSmartMeter

The Factory hides the object-creation process and makes it easier to add new meter types.

## **6. Repository Pattern and OODBMS**

The Repository Pattern provides a clean interface for storing and retrieving persistent energy objects.

Examples include:

- SmartMeterRepository

- EnergyReadingRepository
- AlertRepository
- ConsumerRepository

When a smart meter produces a new reading, the Business Layer validates the reading and passes it to the Repository. The Repository stores the EnergyReading object in the OODBMS. When a dashboard requests historical consumption, the Repository retrieves the required EnergyReading objects and returns them to the Business Layer.

The OODBMS maintains relationships between consumers, meters, readings, alerts, and optimization policies.

## **7. Data Processing Flow**

The smart meter generates an energy reading. The reading is received by the Business Layer, where it is validated and analyzed.

The Domain Layer checks whether the consumption follows the defined rules. If abnormal consumption is detected, the Observer Pattern generates an alert.

The Repository stores the reading and alert information in the OODBMS. When the user opens the dashboard, the Presentation Layer requests information from the Business Layer, which retrieves the required persistent objects through the Repository.

## **8. Scalability and Extensibility**

The architecture can handle additional smart meters and increasing numbers of energy readings by scaling the data access and business services.

Renewable-energy monitoring can be introduced through new domain objects and services. AI-based consumption prediction can be added as a new Strategy without changing the existing monitoring framework.

The use of interfaces, repositories, and loosely coupled components allows future technologies to be integrated with minimal modification.

## **9. Maintainability**

The layered architecture separates presentation, energy analysis, domain rules, and persistence responsibilities. Therefore, modifying the dashboard does not require changing the OODBMS implementation.

The design patterns further improve maintainability:

- **Observer** separates event generation from notification.
- **Strategy** separates optimization algorithms.
- **Factory** separates object creation.
- **Repository** separates persistence operations.

## **Conclusion**

The Smart Energy Management System provides a scalable, extensible, and maintainable solution through object-oriented principles, layered architecture, and design patterns. The Presentation Layer provides monitoring dashboards, the Business Layer performs analysis, the Domain Layer manages energy rules, and the Data Access Layer manages persistence through the OODBMS. The architecture can be extended to support renewable-energy monitoring, smart-grid integration, and AI-based energy prediction without major changes to the existing system.